

Question	Answer	Marks
1(a)	air temperature: K and air pressure: $\text{kg m}^{-1} \text{s}^{-2}$	B1
	scalar only ticked for both air temperature and air pressure	B1
1(b)(i)	$\text{upthrust} = 1.1 \times 9.81 \times (4\pi \times 0.90^3 / 3)$	C1
	$= 33 \text{ N}$	A1
1(b)(ii)	(due to difference in height / depth there is a) difference in pressure between top and bottom (of balloon)	B1
	(due to pressure difference, upwards) force on bottom of balloon is greater (than downwards force on top of balloon, so resultant force is upwards)	B1
1(b)(iii)	$(\Sigma)F = 33 - 19$ $(= 14 \text{ N})$	C1
	$m = 19 / 9.81$ $(= 1.94 \text{ kg})$	C1
	$a = (33 - 19) / (19 / 9.81)$ $= 7.2 \text{ m s}^{-2}$	A1
1(c)	$5 + 3 + (2 \times 4)$ $(= 16\%)$	C1
	absolute uncertainty in $c = 1.8 \times 0.16$ $= (\pm) 0.3$	A1

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